Wenxin Luo

Research Interests: LLM Alignment, Multimodal LLM Security, Machine Learning
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EDUCATION

City University of Hong Kong Jan 2024–Jun 2025
Master of Science in Data Science
GPA: 3.46/4.3

Beijing University of Technology (Project 211) Sep 2019–Jul 2023
Bachelor of Engineering in Software Engineering
Average Score: 85.25/100

PUBLICATIONS

* Equal contribution.

[1] **Wenxin Luo***, Yanzhao Chen*, Ruoxuan Cao, Cuifeng Gao, Ning Liu, Shuai Wang, Zhiyuan Wan, Daoyuan Wu. *GuardAlign: Guardrail-Aware Safety Alignment for Modern LLM Pipelines*. Under review at ICLR 2027. Paper Code

[2] **Wenxin Luo***, Weirui Wang*, Xiaopeng Li*, Weibo Zhou, Pengyue Jia, Xiangyu Zhao. *TAPO: Task-Referenced Adaptation for Prompt Optimization*. 2025 IEEE International Conference on Acoustics, Speech, and Signal Processing. **(ICASSP, CCF-B)** Paper Code

[3] **Wenxin Luo**. *Particle Swarm Optimization Algorithm with Proportional Factor Based on Nash Equilibrium*. 2022 5th International Conference on Advanced Algorithms and Control Engineering. **(ICAACE, EI)** Paper

RESEARCH EXPERIENCE

Lingnan University Sep–Nov 2025; May 2026–Present
Research Assistant Supervisor: Prof. Daoyuan Wu

- Proposed a DPO-style LLM fine-tuning paradigm that incorporates decisions and confidence signals from multiple guardrails into preference optimization, achieving a superior safety–utility trade-off over multiple baselines across diverse benchmarks.
- Conducted a systematic review of MLLM defenses and developed a unified taxonomy for analyzing and comparing diverse defense mechanisms.

City University of Hong Kong Dec 2025–Apr 2026
Research Assistant Supervisors: Prof. Ning Liu, Prof. Daoyuan Wu

- Developed and maintained an academic website to promote seminars and disseminate research. Website

The Hong Kong University of Science and Technology Jun 2025–Sep 2025
Research Assistant Supervisor: Prof. Daoyuan Wu

- Built a safety-alignment dataset capturing adversarial jailbreaks and over-refusal cases, enabling LLMs to correct guardrail errors through guardrail-aware safety alignment.
- Explored methodologies for constructing and evaluating audio-based telecom fraud detection systems, with a focus on near-domain negative samples and LLM-driven red-teaming pipelines.

City University of Hong Kong Jan 2024–May 2025
Student Researcher Supervisor: Prof. Xiangyu Zhao

- Developed an evolution-based method for task-adaptive prompt optimization and designed task-referenced evaluation metrics; the resulting paper was published at ICASSP 2025.
- Built a text classifier using CNN encoding and GCN-based feature propagation; the resulting dissertation was awarded a grade of A. Dissertation

SKILLS & SERVICES

Languages: English (IELTS 6.5); Mandarin (Native)
Programming & Markup: Python, Java, HTML/CSS
Tools & Platforms: Git, Linux, SSH, VS Code, Eclipse
Libraries & Frameworks: PyTorch, Transformers, TensorFlow, NumPy, Pandas, PEFT, Accelerate
Conference Subreviewer: NDSS 2026, ICML 2026, USENIX Security 2026, ICME 2026, CCS 2025